Boston Housing Data

Case Study

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Introduction & Problem Statement

Dataset Background

Source: Harrison & Rubinfeld (1978), MASS package in R

Observations: 506 census tracts in Boston, MA

Predictors: 13 variables (crim, zn, indus, chas, nox, rm, age, dis, rad, tax, ptratio, black, lstat)

Response: medv — Median value of owner-occupied homes ($1000s)

Study Goal

Apply and compare 7 supervised learning models to predict median home value (medv) and identify the best-performing model using out-of-sample MSPE.

Evaluation Approach

Data Split: 80% training / 20% testing (set.seed = 1234)

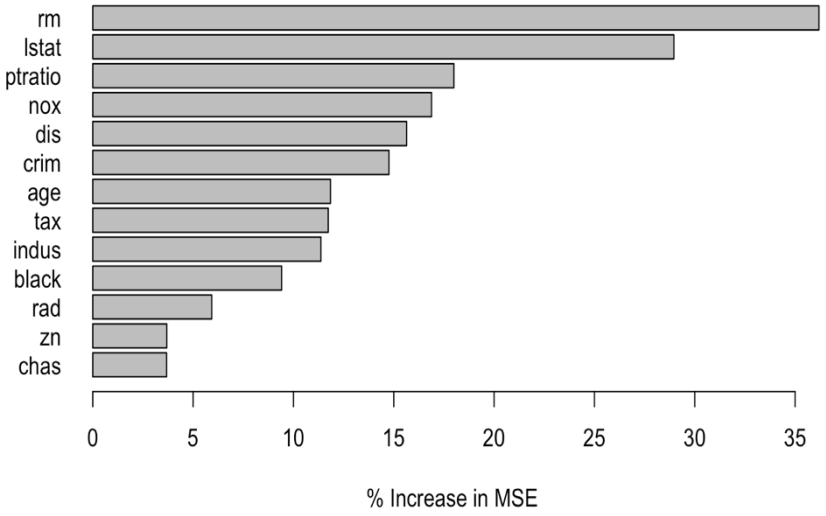
In-Sample: Average Squared Error (ASE)
Out-of-Sample: Mean Squared Prediction Error (MSPE)

Model Summary & Results

Rank	Model	Description	ASE (In-Sample)	MSPE (Out-of-Sample)
1	**Random Forest**	**Ensemble of trees using bootstrap samples and random predictor subsets.**	**2.14**	**11.80★**
2	Boosting	Sequential trees that improve prediction by learning from previous errors.	2.25	12.48
2	Neural Network	Hidden-layer model using scaled variables to capture nonlinear patterns.	4.77	12.48
3	GAM	Flexible model using smooth functions for nonlinear relationships.	8.16	14.79
4	Pruned Regression Tree	Tree model pruned using cross-validated CP to avoid overfitting.	9.73	18.92
5	K-NN	Distance-based model using scaled predictors and optimal k = 2.	6.18	20.57
6	Linear Regression	Stepwise variable selection to identify useful linear predictors.	20.64	28.22

Selective Figures

Figure 1: RF Feature Importance



*The feature importance chart shows that rm and lstat were the most influential predictors, which is consistent with the scatterplots showing a positive relationship for rm and a negative relationship for lstat.

Figure 2: medv vs rm

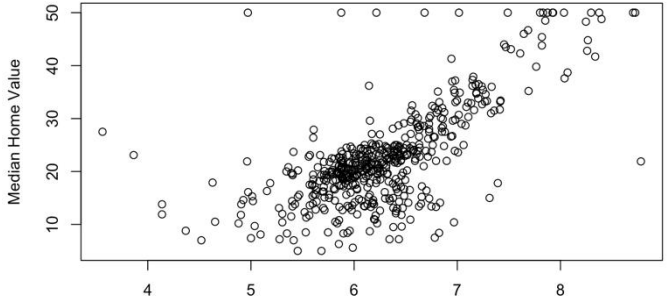
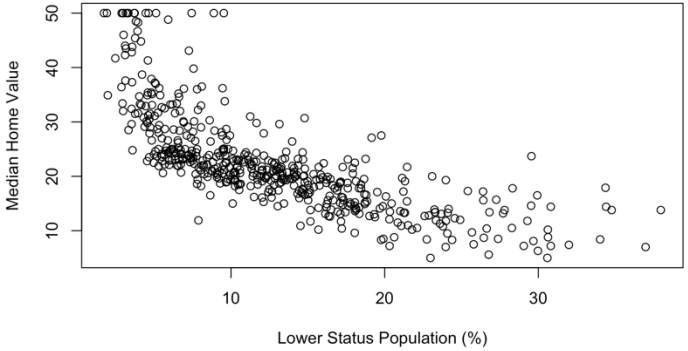


Figure 3: medv vs lstat



Summary & Recommendations

Main Findings

- rm and lstat were the strongest predictors of median home value.
- More rooms were associated with higher home values, while higher lstat was associated with lower home values.
- Nonlinear/ensemble models captured the housing patterns better than simpler models.

Best Model

- **Random Forest** had the lowest out-of-sample MSPE: **11.80**.
- This means it produced the most accurate predictions on the testing data.

Recommendation

- Recommend **Random Forest** as the final predictive model for this case study.
- Use the model to predict median home value, while using feature importance to explain key drivers.
- For interpretation, focus on rm and lstat because they were the most important predictors.

Discussion & Literature Review

Existing Studies

Harrison, D. and Rubinfeld, D.
Showed that housing prices are influenced by structural, neighborhood, and environmental factors; my results support this idea, with rm and lstat emerging as the most important predictors.

Çeh et al. (2018) — ISPRS Int. J. Geo-Inf.
Random Forest outperformed linear regression and SVR on housing data, consistent with our finding of RF superiority (MSPE 11.80 vs. 28.22).

Ashika K. A. (2024)
Found Random Forest to be a top-performing model for the Boston Housing dataset and identified RM and LSTAT as key predictors, which supports my recommendation of Random Forest and my feature importance results.

How Our Results Compare

✓ RF aligns with literature benchmarks as top performer model

✓ lstat and rm as top predictors is consistent with prior hedonic pricing research

✓ Linear regression underperforms, confirming that housing data has nonlinear structure

References

[1] Harrison, D. & Rubinfeld, D.L. (1978). Hedonic prices and the demand for clean air. J. Environ. Economics.
[2] Çeh, M. et al. (2018). Estimating the performance of RF vs. multiple regression for predicting prices. ISPRS Int. J. Geo-Inf.
[3] Ashika K. A. (2024), "Boston Housing Price Prediction & Model Comparison," Kaggle Notebook.

Thank You!